Section Report 2 — WGS: Reads to Info

BIOL 4810

2026-03-12

What you are doing: You are given **raw reads only**. You must run the full WGS workflow and document it in a way that is reproducible and ready to commit to Git.
Submission: Fill this page out, upload the requested screenshots/PDFs into the page, then use **Download as PDF (Print)**.
Notes: You should taking detailed notes about each step, what software you are using, what version of the software (if available), and output files so that someone else could theoretically repeat your exact work.

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0) Required folder structure (make this first)

Inside your course repository, create a folder named something like:

```
sectionreport2_wgs/
```

Inside that folder, create:

```
rawfastqc/
fastp/
trimmedfastqc/
assembly/
quast/
filteredassembly/
checkm2/
prokka/
ani/
tygs/
miningfiles/
```

---

1) FASTQC on raw reads (Galaxy)

In Galaxy 1. Upload your raw reads (make sure to create the paired data set) 2. Run **FastQC** on **raw** reads 3. Open the FastQC HTML report

Download requirement: Download the FastQC HTML report for **either R1 or R2** and save it in: `rawfastqc/`

1A. FastQC version (from Galaxy Job Info):

(Galaxy Version 0.74+galaxy1)

1B. Read length reported:

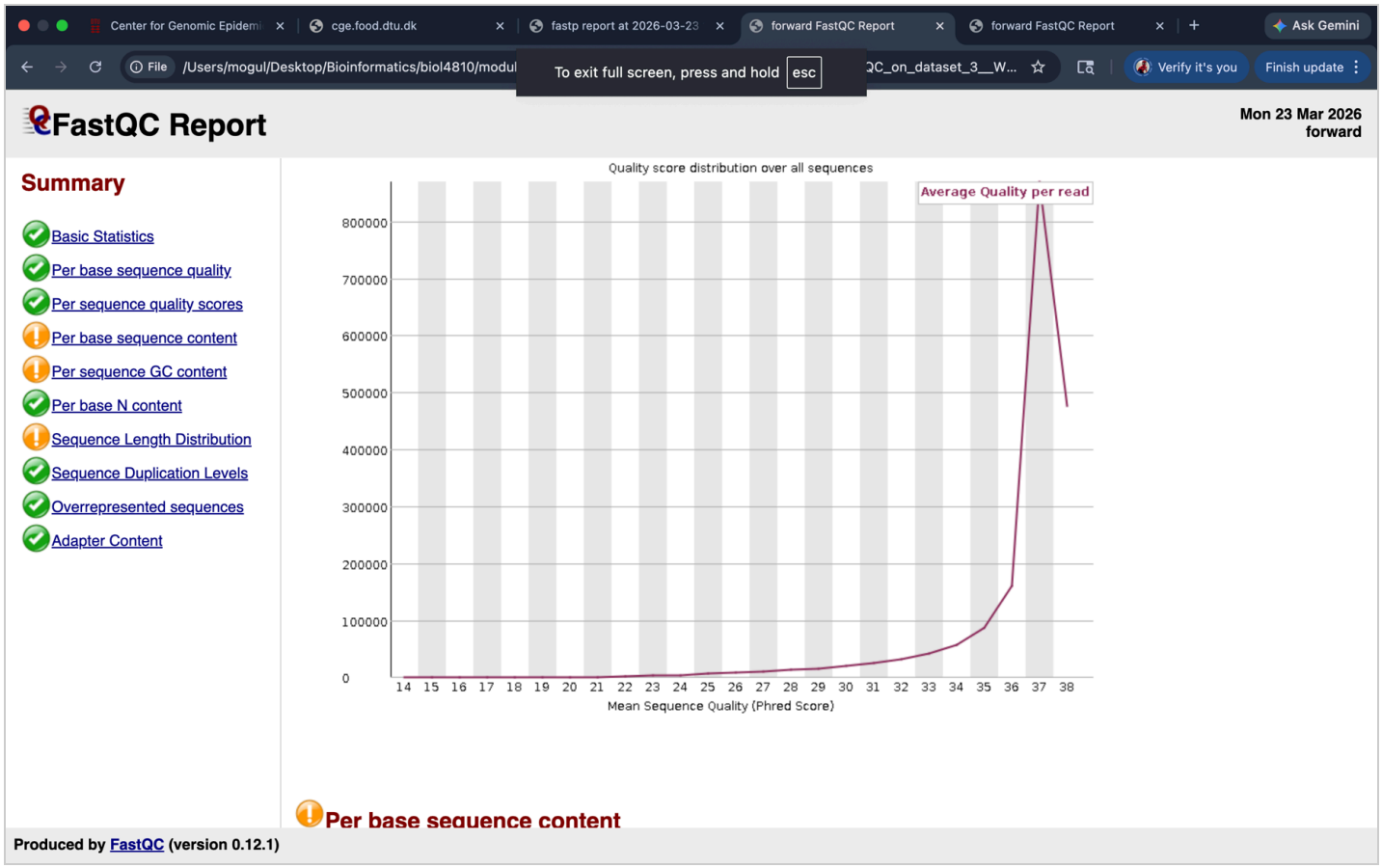
150 bp

1C. One key quality issue you see in raw reads (2–3 sentences):

There were flags for the per sequence GC content. The GC Count per read was higher than the theoretical distribution provided.

Upload screenshot: raw FastQC quality plot (R1 or R2)

Choose File rawfastqcss



Save the downloaded FastQC HTML report in `rawfastqc/` using your naming convention.

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2) fastp trimming/filtering (Galaxy)

In Galaxy 1. Run **fastp** on your raw reads 2. Keep defaults unless your instructor specifies otherwise 3. Make sure the output includes the report

Download requirement: Download the **fastp report HTML** and save it in: `fastp/`

2A. fastp version (from Galaxy Job Info):

Galaxy Version 1.1.0+galaxy0)

2B. What did fastp change? (2–3 sentences)

Fastp changed the mean length down to 147 bp after filtering. And with that, it brought down the total number of reads from 3.689168 M to 3.569176 M, bringing down the number of total bases; Q20, Q30, and Q40 bases; and GC content.

2C. Name one metric that changed from BEFORE → AFTER in the fastp report (e.g., Q30%, GC%, read length distribution) and state how it changed:

GC Content changed from being 65.361302% to 65.195791%.

Upload your fastp HTML report (downloaded from Galaxy)

Choose File fastp_on_d…html.html

Selected file: fastp_on_dataset_3_and_4__HTML_report_html.html

Save your fastp HTML in `fastp/` using your naming convention.

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3) FASTQC on trimmed reads (Galaxy)

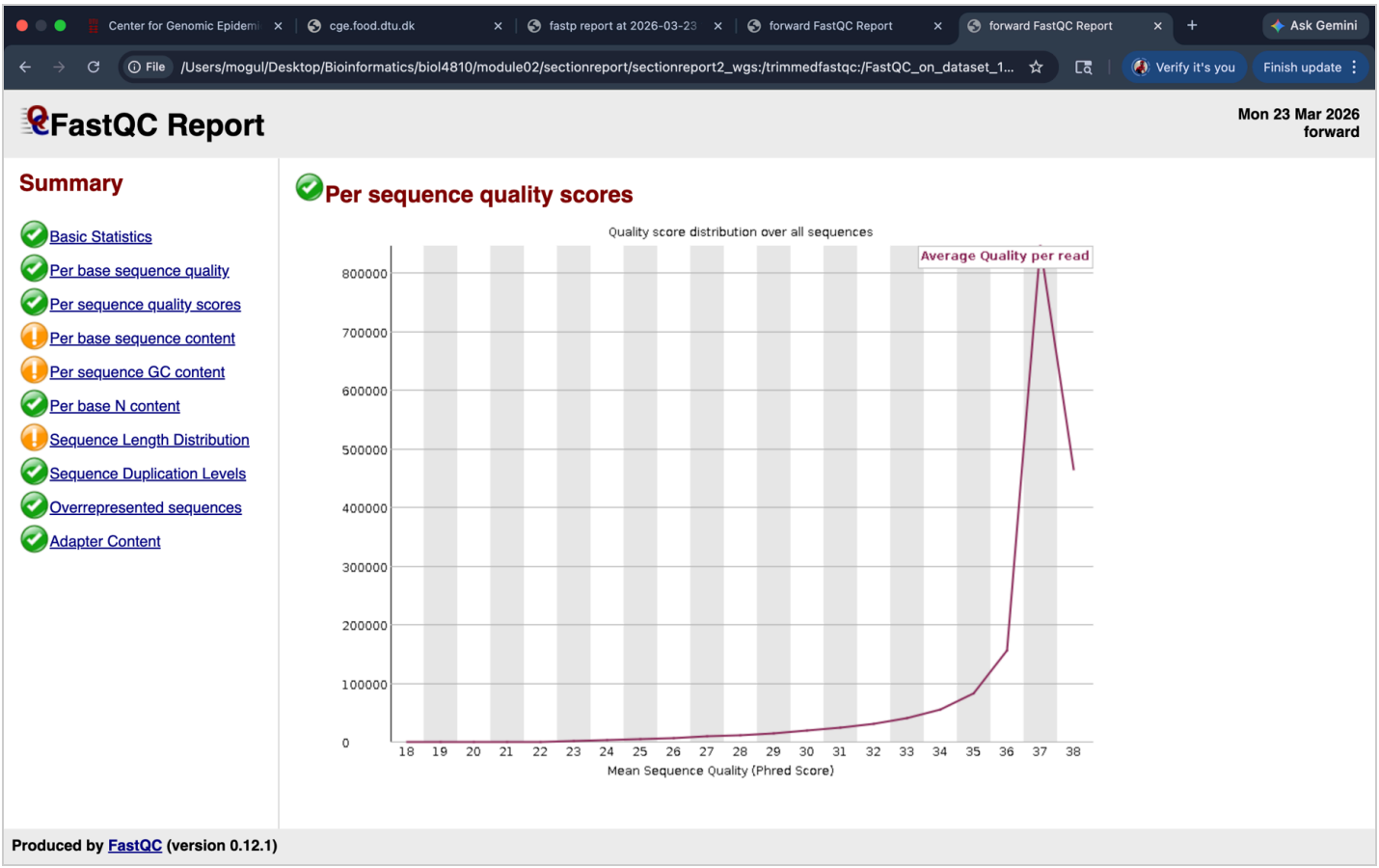
In Galaxy 1. Run **FastQC** on the **trimmed** reads 2. Download the FastQC HTML report for either R1 or R2 and save it in: `trimmedfastqc/`

3A. One key improvement (or remaining issue) after trimming (2–3 sentences):

While the scale has been cut down and the margin for errors are slightly less, the previous flags are still persisting. Quality reads very similarly still.

Upload screenshot: trimmed FastQC quality plot (R1 or R2)

Choose File trimmedfastqcss



Save the downloaded FastQC HTML report in `trimmedfastqc/` using your naming convention.

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4) Assembly (SPAdes in Galaxy)

In Galaxy 1. Run **SPAdes** on your trimmed reads 2. Download the **contigs FASTA** and save it in: `assembly/`

4A. SPAdes version (from Galaxy Job Info):

(Galaxy Version 4.2.0+galaxy0)

4B. Name of your downloaded contigs file (after you renamed it):

unknownsamplecontigs.fasta

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5) Assembly evaluation (QUAST in Galaxy)

In Galaxy 1. Run **QUAST** on your SPAdes contigs 2. Download the **QUAST PDF report only** and save it in: `quast/`

5A. QUAST version (from Galaxy Job Info):

(Galaxy Version 5.3.0+galaxy1)

5B. N50:

135762

5C. L50:

12

5D. Largest contig:

322485

5E. Total assembly length:

4888395

Upload your QUAST PDF report (downloaded from Galaxy)

Choose File quastonspades.pdf

Selected file: quastonspades.pdf

Save the QUAST PDF in `quast/` using your naming convention.

5F. Based on QUAST, is your assembly fragmented? Use at least 2 metrics to justify (2–4 sentences):

The assembly is fragmented, but not terribly fragmented. Evidence of this is in the number of contains that are seen in the QUAST report: 66.

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6) Genome quality (CheckM2 in Galaxy)

6A) Filter FASTA (Galaxy)

To reduce tiny contigs, create a filtered assembly for work going forward:

In Galaxy 1. Use **Filter FASTA** 2. Input: your SPAdes contigs FASTA 3. Filter by minimum contig length: **500 bp** 4. Save the filtered contigs as a new dataset in `filteredcontigs`

6A. Confirm your filter rule (one line):

Kept it at minimum 500 bp

6B) In Galaxy

1. Run **CheckM2** on your assembly (contigs)
2. Download the CheckM2 report/output and save it in: `checkm2/`

6B. CheckM2 version (from Galaxy Job Info):

(Galaxy Version 1.1.0+galaxy0)

6C. Completeness (%):

100.0

6D. Contamination (%):

0.62

6E. Interpret these numbers: is this “good enough” for downstream annotation? Why? (2–3 sentences)

From these numbers, I'd gauge these as "good enough". With minimal contamination at 0.62% and a completeness of 100%, this seems viable for downstream annotation.

Folder requirement: Put the downloaded CheckM2 report/output in `checkm2/` with your naming convention.

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7) Prokka annotation (Galaxy)

In Galaxy 1. Run **Prokka** 2. Input: your **filtered** contigs (≥500 bp) 3. Change at least one setting from default as instructed (record it below) 4. Run Prokka

7A. Prokka version (from Galaxy Job Info):

(Galaxy Version 1.14.6+galaxy1)

7B. What did you change from Prokka defaults? (1–2 sentences)

Said YES to forced GenBank Compliance.

Download requirements (Prokka): download and save these into `prokka/` - `.txt` (summary) - `.gbk` (GenBank format annotation) - `.ffn` (nucleotide sequences of features)

7C. Paste your Prokka summary lines (from the .txt) and describe why this does or does not look like a typical bacterial genome:

rRNA: 4
tRNA: 71
tmRNA: 1

This looks like a typical bacterial genome because of the number of bases it has lines up with what is typical for bacteria (3-5 million bp)

7D. How many hypothetical proteins does your organism have? What percent of CDS are hypothetical?

2071 hypothetical proteins, 51.24%

rRNA feature check (from Prokka outputs)

Open your Prokka outputs and locate the rRNA entries (typically in the `.ffn` and/or `.tbl` / `.gff` depending on Galaxy wrapper).

7E. Paste the description/header lines for 23S, 16S, and 5S rRNA (three lines total):

>COLHKAPM_02454 23S rRNA (uracil(1939)-C(5))-methyltransferase RlmD
>COLHKAPM_04500 16S ribosomal RNA
>COLHKAPM_03185 5S ribosomal RNA

7F. Paste the full 16S rRNA FASTA sequence (header + sequence):

```
>COLHKAPM_04500 16S ribosomal RNA
TGAAGAGTTTGATCCTGGCTCAGAGTGAACGCTGGCGGTAGGCCTAACACATGCAAGTCG
AACGGCAGCACAGGAGAGCTTGCTCTCTGGGTGGCGAGTGGCGGACGGGTGAGGAATACA
TCGGAATCTACTTTTTCGTGGGGGATAACGTAGGGAAACTTACGCTAATACCGCATACGA
CCTACGGGTGAAAGCAGGGGATCTTCGGACCTTGCGCGATTGAATGAGCCGATGTCGGAT
TAGCTAGTTGGCGGGGTAAAGGCCCACCAAGGCGACGATCCGTAGCTGGTCTGAGAGGAT
GATCAGCCACACTGGAACTGAGACACGGTCCAGACTCCTACGGGAGGCAGCAGTGGGGAA
TATTGGACAATGGGCGCAAGCCTGATCCAGCCATACCGCGTGGGTGAAGAAGGCCTTCGG
GTTGTAAAGCCCTTTTGTTGGGAAAGAAATCCAGCTGGTTAATACCCGGTTGGGATGACG
GTACCCAAAGAATAAGCACCGGCTAACTTCGTGCCAGCAGCCGCGGTAATACGAAGGGTG
```

8) 16S species ID using BLAST (NCBI)

Use your 16S rRNA FASTA from above.

8A. Top hit species name + percent identity + query coverage:

Stenotrophomonas maltophilia + 100.00% + 100%

8B. Based on 16S alone, how confident are you in species-level ID? Why? (2–3 sentences)

While the percent identity and the query coverage are at 100% and should give good confidence, my confidence for species ID isn't completely perfect. This is because a lot of bacteria share conserved regions of 16s rRNA, so this identity could be shared with other bacterial species.

9) ANI (EZBioCloud) — two comparisons

Use the ANI tool and compare your assembly to **two reference genomes** that you choose (based on your best guess from 16S/Prokka/TYGS). Open the site here (new tab): EZBioCloud ANI (https://www.ezbiocloud.net/tools/ani)

Download requirement: download the two reference genomes you used and save them in `ani/` with clear names.

9A. Reference genome #1 (species/strain):

Stenotrophomonas maltophilia

9B. ANI % vs reference #1:

98.10%

9C. Reference genome #2 (species/strain):

Stenotrophomonas pavanii

9D. ANI % vs reference #2:

91.12%

9E. Interpret your ANI: do you meet a typical same-species ANI threshold? What conclusion do you draw? (2–4 sentences)

The range for a typical same-species ANI threshold is 95-96%, so I'm pretty confident in the identity as S. maltophilia. As for S. pavanii, the ANI value being within 90-95% at least proves that the unknown is within the Strenotrophomonas genus, so that helps build confidence in its identity.

10) TYGS phylogeny (use filtered contigs for this report)

Run TYGS using your filtered contigs (≥500 bp). Export/download the tree (.phy file) and view in iTOL. Make sure that you have displayed the bootstrap values. Export your tree as a .pdf file. The .phy and .pdf should be in your `tygs/` folder

In your own words: what kinds of evidence does TYGS use to place your genome? Did you need to generate a proteome tree or was the genome tree sufficient? Did it correspond to what you found with your 16S BLAST results.

Unfortunately, I wasn't able to generate a proteome tree due to the TYGS not providing me with a tree. however, TYGS would have been helpful by placing our genome amongst other organism entries within its database and providing a clearer picture about what the unknown organism could possibly be.

Upload your TYGS/iTOL tree as a PDF

Choose File no file selected

11) Genome mining (Day 14 web tools) — ALL REQUIRED

You must run **all** tools below and include a **screenshot of results** for each. Save screenshots in: `miningfiles/` with clear names.

Also: record the **version/date/database** if the tool reports it (many do on the results page or in the footer).

11A) CARD / RGI (AMR genes)

Open: CARD / RGI (https://card.mcmaster.ca/analyze/rgi)

11A1. CARD/RGI version + database/date info (copy from results page if shown):

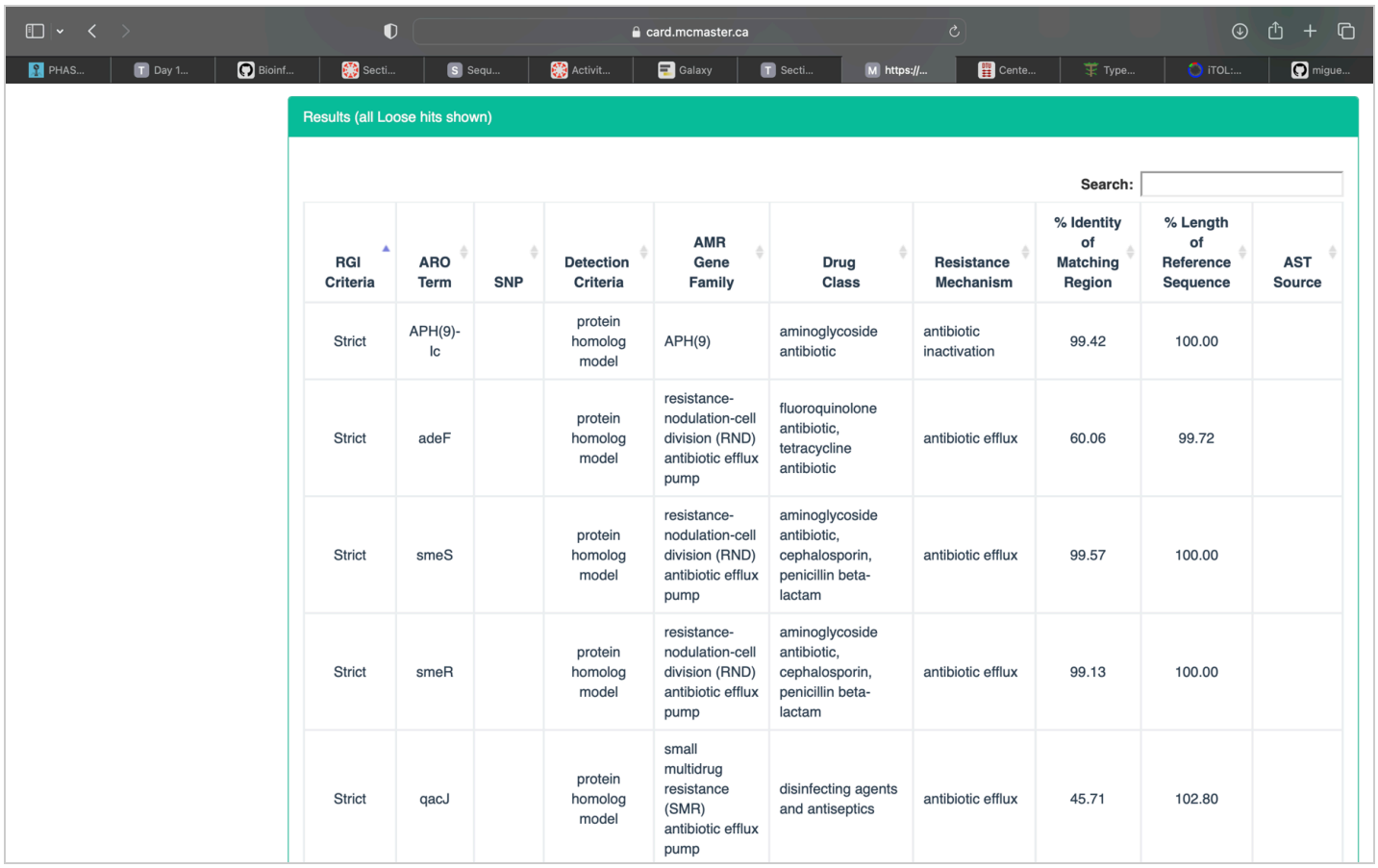
Web portal - RGI 6.0.5, CARD 4.0.1

11A2. Key AMR findings (2–4 sentences). Include at least one gene + evidence type (e.g., perfect/strict/loose):

There was an AMR finding for APH (9) which has the ability of antibiotic inactivation. It has a %identity of 99.42% and %length of a reference sequence of 100.00% pretty strong evidence. We also find that there are 5 strict hits for the RGI criteria.

Upload screenshot: CARD/RGI results

Choose File amrss



RGI Criteria	ARO Term	SNP	Detection Criteria	AMR Gene Family	Drug Class	Resistance Mechanism	% Identity of Matching Region	% Length of Reference Sequence	AST Source
Strict	APH(9)-Ic		protein homolog model	APH(9)	aminoglycoside antibiotic	antibiotic inactivation	99.42	100.00	
Strict	adeF		protein homolog model	resistance-nodulation-cell division (RND) antibiotic efflux pump	fluoroquinolone antibiotic, tetracycline antibiotic	antibiotic efflux	60.06	99.72	
Strict	smeS		protein homolog model	resistance-nodulation-cell division (RND) antibiotic efflux pump	aminoglycoside antibiotic, cephalosporin, penicillin beta-lactam	antibiotic efflux	99.57	100.00	
Strict	smeR		protein homolog model	resistance-nodulation-cell division (RND) antibiotic efflux pump	aminoglycoside antibiotic, cephalosporin, penicillin beta-lactam	antibiotic efflux	99.13	100.00	
Strict	qacJ		protein homolog model	small multidrug resistance (SMR) antibiotic efflux pump	disinfecting agents and antiseptics	antibiotic efflux	45.71	102.80	

Save as something like `SampleID_card_rgi.png` in `miningfiles/` .

11B) ResFinder (AMR genes)

Open: ResFinder (https://genepi.food.dtu.dk/resfinder)

11B1. ResFinder version/database info (copy from results page if shown):

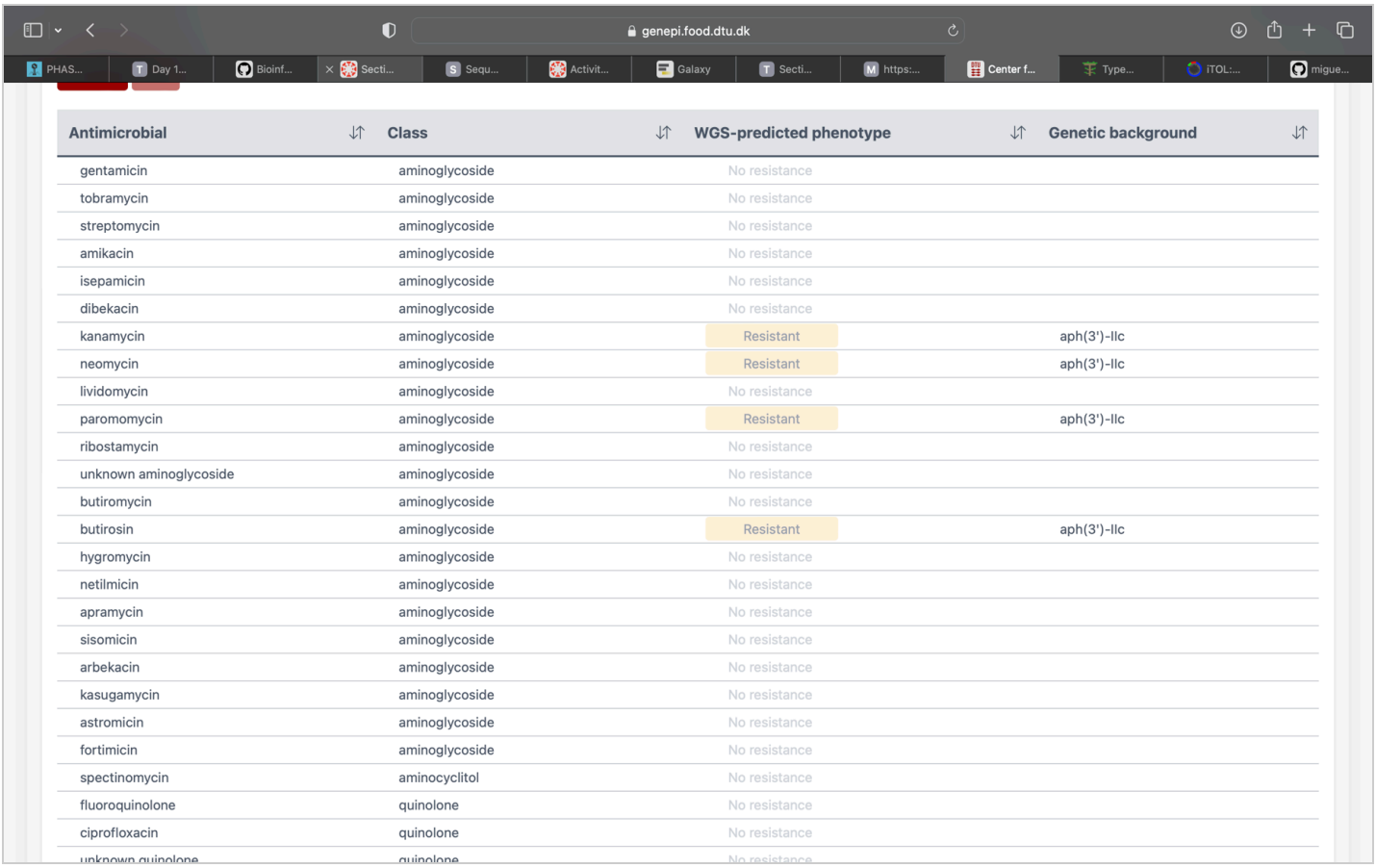
4.7.2

11B2. Key findings (2–4 sentences). Do ResFinder and CARD agree? If not, why might they differ?

They agree in that the founded resistance was to aminoglycoside class antibiotics. However, the 5 strict hits in CARD differ from the 4 WGS-predicted phenotypes in ResFinder.

Upload screenshot: ResFinder results

Choose File resfinderss



Antimicrobial	Class	WGS-predicted phenotype	Genetic background
gentamicin	aminoglycoside	No resistance	
tobramycin	aminoglycoside	No resistance	
streptomycin	aminoglycoside	No resistance	
amikacin	aminoglycoside	No resistance	
isepamicin	aminoglycoside	No resistance	
dibekacin	aminoglycoside	No resistance	
kanamycin	aminoglycoside	Resistant	aph(3')-IIc
neomycin	aminoglycoside	Resistant	aph(3')-IIc
lividomycin	aminoglycoside	No resistance	
paromomycin	aminoglycoside	Resistant	aph(3')-IIc
ribostamycin	aminoglycoside	No resistance	
unknown aminoglycoside	aminoglycoside	No resistance	
butiromycin	aminoglycoside	No resistance	
butirosin	aminoglycoside	Resistant	aph(3')-IIc
hygromycin	aminoglycoside	No resistance	
netilmicin	aminoglycoside	No resistance	
apramycin	aminoglycoside	No resistance	
sisomicin	aminoglycoside	No resistance	
arbekacin	aminoglycoside	No resistance	
kasugamycin	aminoglycoside	No resistance	
astromicin	aminoglycoside	No resistance	
fortimicin	aminoglycoside	No resistance	
spectinomycin	aminocyclitol	No resistance	
fluoroquinolone	quinolone	No resistance	
ciprofloxacin	quinolone	No resistance	

Save as `SampleID_resfinder.png` in `miningfiles/`.

11C) PathogenFinder 2.0 (pathogenicity prediction)

Open: PathogenFinder (https://genepi.food.dtu.dk/pathogenfinder)

11C1. PathogenFinder version/database info (copy from results page if shown):

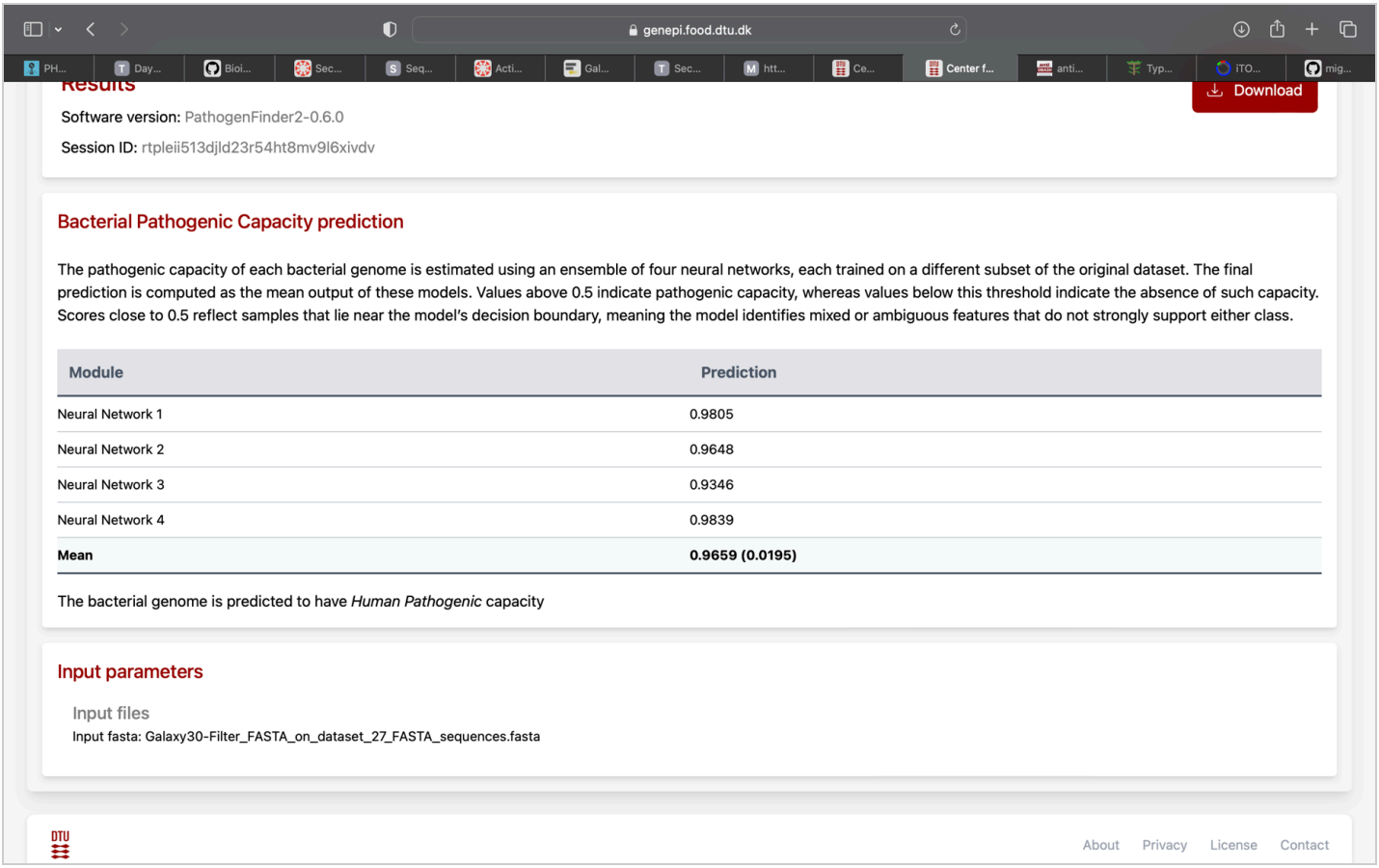
0.6.0

11C2. Report the predicted probability/label and interpret it (2–4 sentences):

The mean predicted probability is 0.9659 (0.0195). This means that it is predicted to have the ability to be a human pathogen based on its sequence.

Upload screenshot: PathogenFinder results

Choose File pathfinderss



Save as `SampleID_pathogenfinder.png` in `miningfiles/`.

11D) VFDB virulence factors via **ABRicate** (Galaxy)

The VirulenceFinder/VFDB web outputs can be inconsistent, so for this report you will run VFDB searches **inside Galaxy** using **ABRicate**.

In Galaxy (ABRicate): 1. Search tools for **ABRicate** 2. **Input:** your **Prokka GenBank file** (`.gbk`) 3. Under**Advanced**, **Database:** choose **VFDB** 4. Run ABRicate 5. Download the **tabular** results file (`.tabular`) and save it in `miningfiles/` using your naming convention.

11D1. ABRicate version + VFDB database version (from Galaxy Job Info):

(Galaxy Version 1.0.1)

11D2. How many VFDB hits did you get (from the results table/summary)?

2

11D3. List your top 3 VFDB hits (gene name + brief description/function). If you have fewer than three, list all. If you have none, you can put NA:

aph(3')-IIc_1 + inactivates amino glycoside antibiotics through phosphorylation

blaL1_3 + hydrolyzes broad spectrum Beta-Lactum antibiotics

Upload your ABRicate VFDB results table (tabular)

Choose File prokkaabricate.tabular

Selected file: prokkaabricate.tabular

Save as `SampleID_abricate_vfdb.tabular` in `miningfiles/` .

11E) antiSMASH (biosynthetic gene clusters)

Open: antiSMASH (https://antismash.secondarymetabolites.org/)

11E1. antiSMASH version/database info (copy from results page if shown):

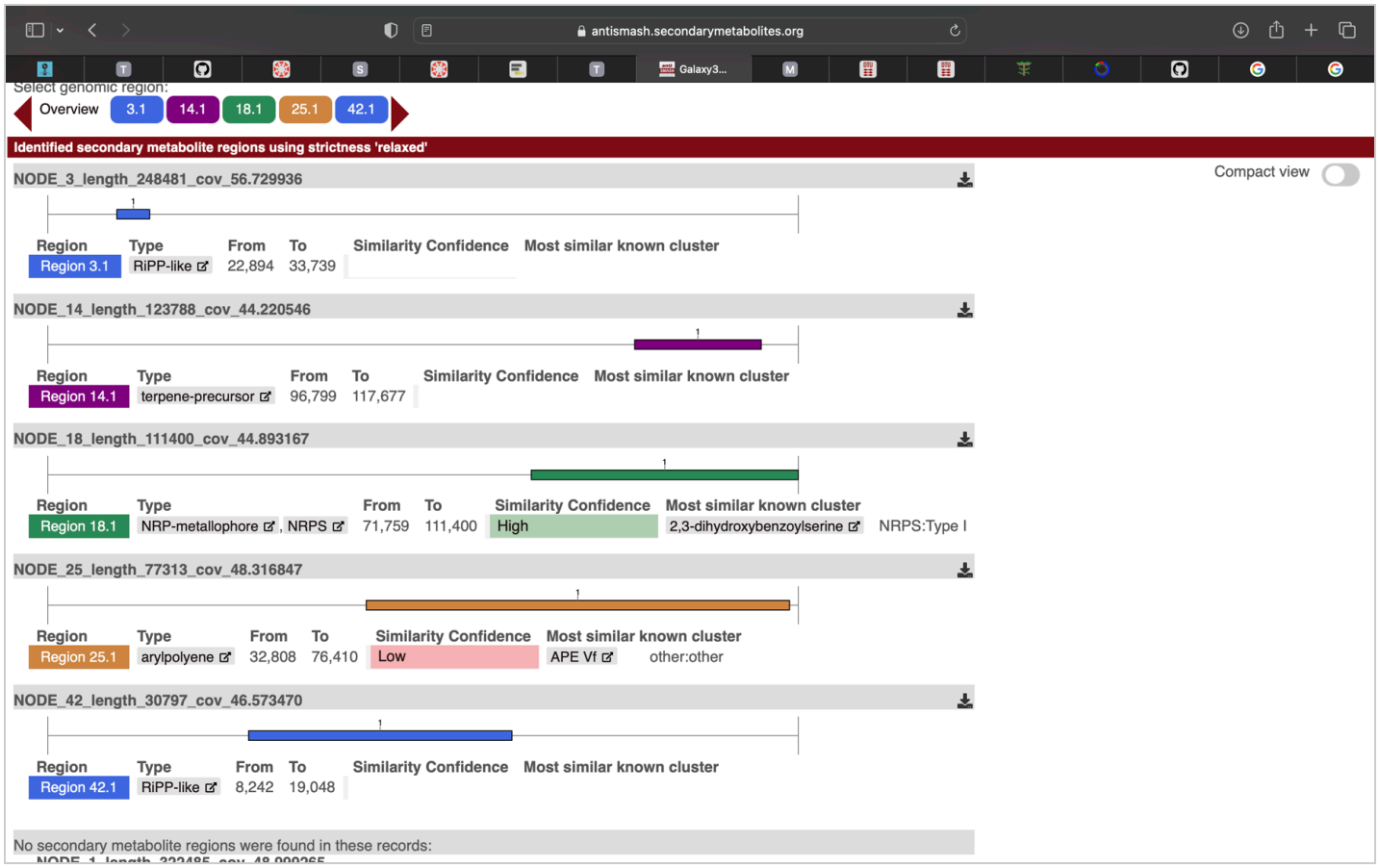
8.0

11E2. Key BGC finding(s) (2–4 sentences). Include at least one cluster type and whether it matches a known cluster:

There were 5 regions identified using antiSMASH. Of these, there are two cluster types that match a respective known cluster: NRP - metallophore with a high similarity confidence for 2,3-dihydroxybenzoylserine; and arylpolyene with a low similarity confidence for APE Vf.

Upload screenshot: antiSMASH results

Choose File antismashss



Save as `SampleID_antismash.png` in `miningfiles/`.

11F) IslandViewer (genomic islands)

Open: IslandViewer (https://www.pathogenomics.sfu.ca/islandviewer/)

11F1. IslandViewer version/database info (copy from results page if shown):

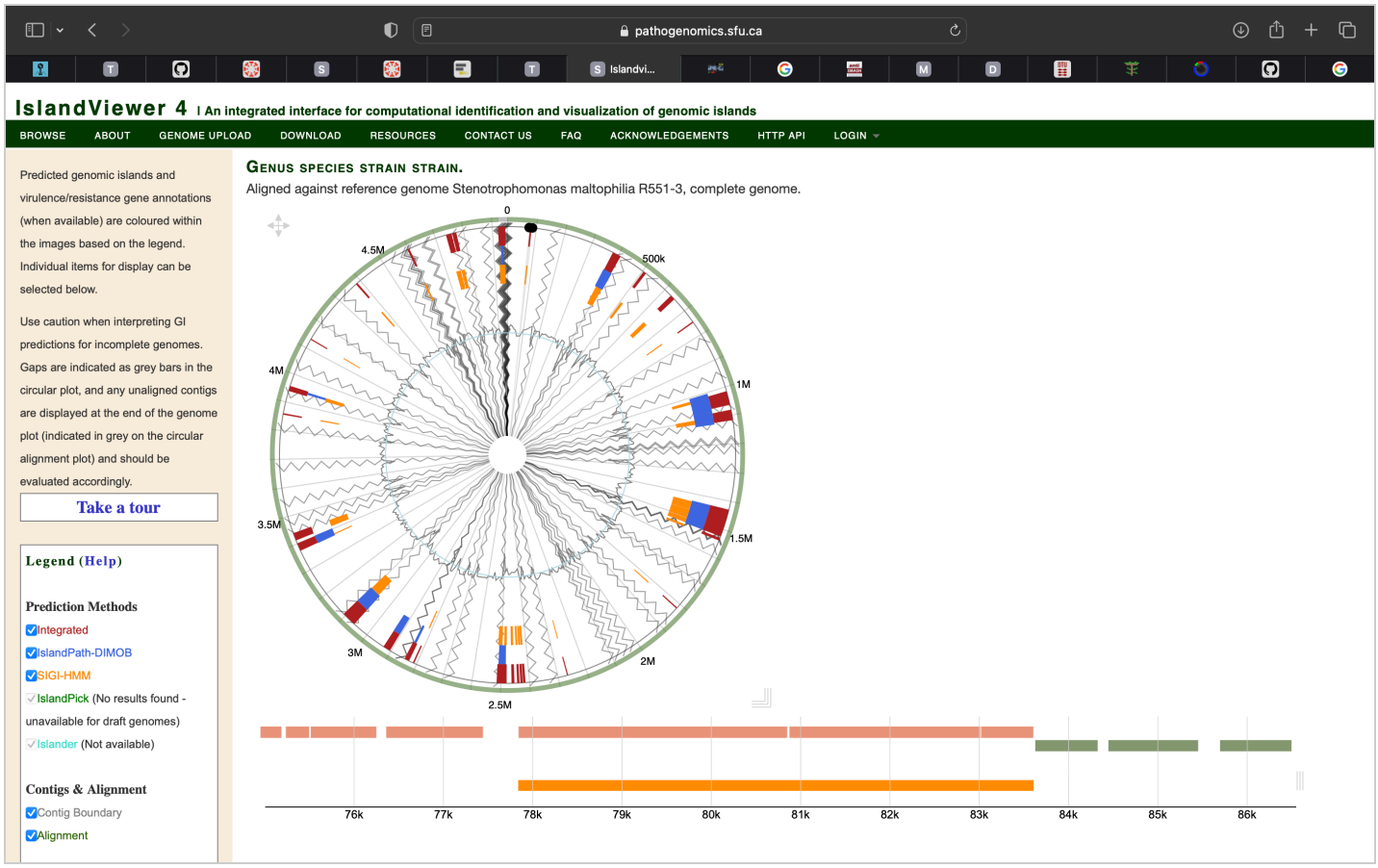
website: https://www.pathogenomics.sfu.ca/islandviewer/results/QG2Ip1FUqaEnoeuwMJqxaY/

11F2. Key island finding(s) (2–4 sentences). How many predicted islands? Any notable features (mobility genes, tRNA insertion sites, etc.)?

There were about 10 predicted islands. There is a feature within called Isoleucine--tRNA ligase which is a factor that aids in translation by activating isoleucine.

Upload screenshot: IslandViewer results

Choose File islandviewerss



Save as `SampleID_islandviewer.png` in `miningfiles/`.

11G) CRISPRCasFinder (CRISPR arrays + Cas genes)

Open: CRISPRCasFinder (https://crisprcas.i2bc.paris-saclay.fr/)

11G1. CRISPRCasFinder version/database info (copy from results page if shown):

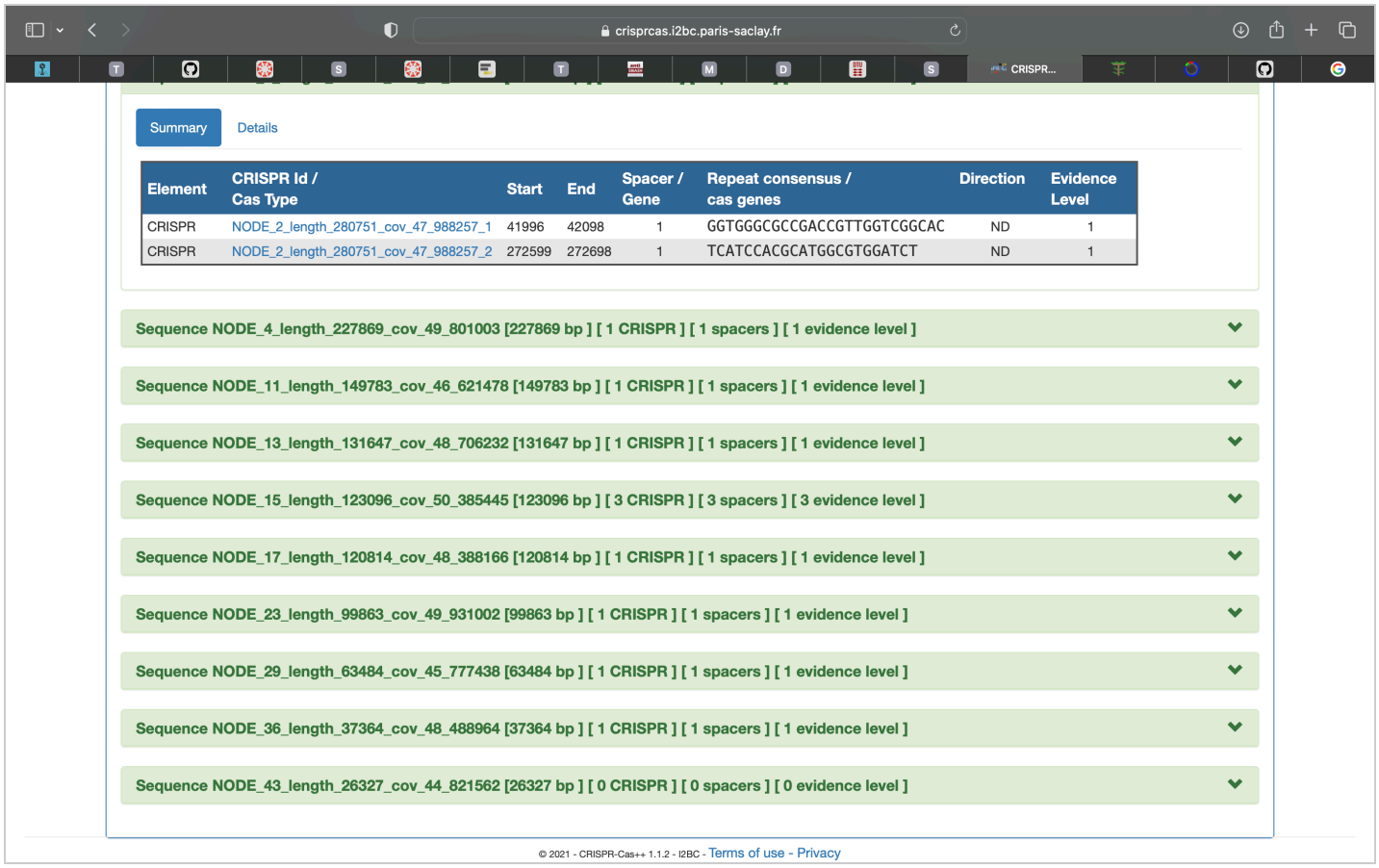
CRISPR-Cas++ 1.1.2 - I2BC

11G2. Key CRISPR/Cas findings (2–4 sentences). Did you find CRISPR arrays? Cas genes? What confidence level?

There were several CRISPR arrays found. However, when viewing all of them they all have evidence levels of 1, so the confidence levels aren't very high throughout despite the list of findings.

Upload screenshot: CRISPRCasFinder results

Choose File crisprcasfinderss



Save as `SampleID_crisprcasfinder.png` in `miningfiles/` .

11H) Synthesis across genome mining tools

11H1. In 6–10 sentences, synthesize what genome mining suggests about this organism (resistance potential, virulence potential, mobile elements, secondary metabolism, etc.). Point to at least one result from each tool:

CARD and ResFinder suggests that this organism has some resistance against aminoglycoside antibiotics. It is also suggested that this organism has the potential ability to be a human pathogen according to PathogenFinder. As for the potential in the organism's virulence, the factors determined from ABRicate suggests it has some potential through two factors from its readings. antiSMASH gave us suggestions about the possible metabolic pathways that the organism has by giving us a comparatively known Custer to a region within the sequence. There is also some evidence to suggest that the organism has been previously attacked by bacteriophages and has adapted accordingly as evidenced by CRISPRCasFinder. The organism also shares a lot of "island" functions as proven by IslandViewer, such as possible virulence factors, possible

12) Final synthesis (your conclusion)

12A. What is your best-supported species ID? Use at least 3 evidence types (16S, ANI, TYGS, CheckM2, Prokka) and explain remaining uncertainty (8–12 sentences):

The best-supported species ID is for Stenotrophomonas maltophilia. The 16S rRNA gathered from prokka and fed into BLAST supports this with a 100% query coverage and a 100% identity match. However, confidence from this can be questioned due to shared 16S rRNA amongst different bacteria species. This was mitigated with ANI through EZBioCloud and the two reference genomes used. The ANI percentages for both make a combined understanding that the organism was highly likely to be Stenotrophomonas maltophilia, and also ensures confidence through an ANI percentage value for a different species under the same genus which proves the similarities in genus. After running checkM2 on it with the data reading contamination percentage and a 100% completeness, confidence in annotation can be assured with minimal room for error. I have some remaining uncertainty due to a certain lack of comparison to an extensive database, one that could have been mitigated if TYGS had given me the data. If it did, that would have given

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